Photonic Quantum Error Correction from Geometric Stability Principles in SEFI/GWFM

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Photonic quantum information platforms are highly susceptible to loss, dephasing, mode mixing, and spectral drift. Conventional photonic quantum error correction (QEC) strategies provide effective code constructions, but they do not yet rest on a single physical principle that explains why errors arise and how correction should be structured.

Here, we introduce a geometric framework for photonic QEC derived from the Single Entity Field Interpretation (SEFI) and the Geometric Worldline Field Model (GWFM). In this model, photonic errors correspond to geometric displacements on stability surfaces, and correction corresponds to restoring the sovereignty of the encoded field. We derive stabilizer structures, geometric syndrome relations, and error-surface topology for photonic modes, and we outline experimental implementations based on integrated photonics and weak or quantum-nondemolition (QND) measurement. We further show how SEFI/GWFM offers a geometric interpretation of SPDC-based quantum randomness production and Bloch-sphere CHSH Bell-inequality behavior. This framework provides a physically grounded foundation for photonic QEC and suggests new bosonic-code designs with intrinsic stability advantages.

I. INTRODUCTION

Photonic quantum information processing has emerged as a leading platform for quantum communication, sensing, and scalable quantum computing. Photons offer long coherence times, room-temperature operation, and compatibility with integrated photonics. However, photonic systems are susceptible to distinct error channels, including loss, mode deformation, spectral wandering, and imperfect interference.

Existing photonic QEC approaches—including cat codes, binomial codes, and GKP codes—provide useful mathematical tools for error suppression, but they do not arise from a single physical principle that governs error formation or dictates the architecture of correction.

SEFI[1] and GWFM[2] provide such a principle. SEFI formalizes layered field identity through FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, and WARP EXPRESSION. GWFM provides the geometric worldline foundation for field propagation, curvature, torsion, and stability.

This paper develops a photonic QEC model grounded in these geometric principles and shows that SEFI/GWFM also provides geometric interpretations of SPDC-based QRNG and CHSH Bell-inequality behavior.

II. SEFI/GWFM STABILITY GEOMETRY FOR PHOTONIC MODES

A. Field Origin and Photonic Identity

In SEFI, FIELD ORIGIN defines the primitive geometric identity of a field. For a photonic mode, the origin layer determines the baseline worldline geometry, spectral identity, polarization geometry, and curvature constraints. Stability is treated as a primitive property through the bound

$$\left\| \frac{d^2 r}{dt^2} \right\| \leq S_0, \tag{1}$$

where S_0 is the sovereignty-bound acceleration limit[1].

B. Field Sovereignty and Stability Surfaces

FIELD SOVEREIGNTY ensures invariance under allowed geometric transformations:

$$F(T(r(t))) = F(r(t)). \tag{2}$$

This defines a stability surface: a geometric manifold representing the allowed transformations of a photonic mode.

Errors correspond to displacements orthogonal to this surface.

C. GWFM Worldline Geometry

GWFM defines curvature as

$$K(t) = \frac{\|r'(t) \times r''(t)\|}{\|r'(t)\|^3}, \tag{3}$$

and treats propagation as a geometric worldline subject to curvature and torsion constraints.

Loss corresponds to worldline termination; dephasing corresponds to curvature-induced torsion; mode mixing corresponds to displacement into neighboring stability surfaces.

III. GEOMETRIC PHOTONIC ERROR SURFACES

A. Loss Errors

Loss is modeled as collapse of the worldline into a boundary region of the stability surface.

B. Dephasing Errors

Dephasing corresponds to geometric torsion of the stability surface.

C. Mode-Mixing Errors

Mode mixing corresponds to displacement into a neighboring stability surface.

D. Spectral Wandering

Spectral drift corresponds to translation along the spectral axis of the stability manifold.

IV. GEOMETRIC STABILIZERS AND SYNDROME EXTRACTION

A. Stabilizer Geometry

A stabilizer is defined as a geometric constraint that preserves the sovereignty of the encoded field. For photonic modes, stabilizers correspond to curvature constraints, torsion constraints, spectral-position constraints, and polarization-surface constraints.

B. Syndrome Geometry

Weak or QND photonic measurements can sample curvature deviation, torsion deviation, spectral displacement, and polarization rotation. These measurements define a geometric syndrome vector.

V. SEFI/GWFM INTERPRETATION OF SPDC-BASED QRNG

Spontaneous parametric down-conversion (SPDC) produces photon pairs with strong temporal and spatial correlations due to energy and momentum conservation[3,4]. In non-collinear, degenerate SPDC, the photons appear on an annular ring whose geometry reflects the underlying momentum-matching conditions[5]. Recent work has shown that sectioning this ring enables high-bit-rate quantum random number generation (QRNG) by exploiting correlations between diametrically opposite regions[6].

Within SEFI, the annular spatial distribution corresponds to a stability surface defined by FIELD ORIGIN and FIELD SOVEREIGNTY, where diametrically opposite photon positions represent sovereign geometric symmetries of the pair field[1]. The primitive stability constraint ensures that the worldline curvature of each photon remains bounded, producing the highly regular annular geometry observed experimentally.

GWFM models the propagation of each photon as a geometric worldline constrained by curvature and torsion, providing a physical basis for the strong temporal correlations observed between spatially separated sections of the SPDC ring[2]. Randomness extraction corresponds to sampling warp-expression variations across distinct worldline bundles, enabling multi-bit QRNG without beam-splitter-based probabilistic routing.

VI. SEFI/GWFM INTERPRETATION OF CHSH BELL-INEQUALITY GEOMETRY

Recent experimental work has demonstrated CHSH Bell-inequality violations across the full Poincaré–Bloch sphere, including both Bell and non-Bell maximally entangled states[7]. These results show that CHSH violation is strongly dependent on the geometric relationship between the polarization bases used for each photon.

Within SEFI, polarization-entangled photon pairs occupy distinct regions of a stability surface defined by FIELD ORIGIN and FIELD SOVEREIGNTY[1]. Sovereignty requires that allowed geometric transformations preserve field identity,

$$F(T(r(t))) = F(r(t)), \tag{4}$$

which implies that CHSH violation occurs only when measurement bases correspond to transformations lying within the sovereignty-preserving submanifold of the stability surface.

GWFM provides a complementary interpretation: polarization evolution is treated as a worldline on the Poincaré–Bloch sphere, with curvature and torsion determining the geometric relationship between measurement bases. Bell states correspond to worldlines with minimal torsion, enabling CHSH violation across a wide range of basis choices, whereas non-Bell states correspond to worldlines with higher torsion, producing basis-dependent suppression of violation.

VII. EXPERIMENTAL IMPLEMENTATION

A. Integrated Photonics

Waveguide-based photonic circuits can implement geometric stabilizers, weak or QND sampling, spectral-geometry control, and mode-geometry correction.

B. Weak/QND Measurement

QND detectors provide geometric sampling without worldline termination.

C. Photonic Syndrome Extraction

Syndromes are extracted through homodyne sampling, spectral interferometry, polarization tomography, and mode-shape reconstruction.

D. Correction Operations

Correction corresponds to geometric restoration: spectral translation, phase-geometry correction, mode-shape reformation, and polarization-surface restoration.

VIII. CONCLUSION

We have introduced a geometric framework for photonic quantum error correction derived from SEFI/GWFM stability principles. Photonic errors correspond to geometric displacements on stability surfaces, and correction corresponds to restoring field sovereignty. We further show that SEFI/GWFM provides geometric interpretations of SPDC-based QRNG and CHSH Bell-inequality behavior. This establishes a physically grounded foundation for photonic QEC and opens a new direction for geometric quantum information theory.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

No datasets were generated or analyzed during the current study.

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